Chapter 20

Area Under Curve

Only One Option Correct Type Questions

1. Let the straight line $x = b$ divide the area enclosed by $y = (1 - x)^2$, $y = 0$ and $x = 0$ into two parts $R_1(0 \le x \le b)$ and $R_2(b \le x \le 1)$ such that $R_1 - R_2 = \frac{1}{4}$. Then b equals **[IIT-JEE-2011 (Paper-1)]**

   (A) $\frac{3}{4}$ (B) $\frac{1}{2}$

   (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

2. The area enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ is **[JEE (Adv)-2013 (Paper-1)]**

   (A) $4(\sqrt{2} - 1)$ (B) $2\sqrt{2}(\sqrt{2} - 1)$

   (C) $2(\sqrt{2} + 1)$ (D) $2\sqrt{2}(\sqrt{2} + 1)$

3. The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the points P, Q and the parabola at the points R, S. Then the area of the quadrilateral $PQRS$ is **[JEE (Adv)-2014 (Paper-2)]**

   (A) 3 (B) 6

   (C) 9 (D) 15

4. Area of the region $\left\{(x, y) \in \mathbb{R}^2 : y \ge \sqrt{|x + 3|},\ 5y \le x + 9 \le 15\right\}$ is equal to **[JEE (Adv)-2016 (Paper-2)]**

   (A) $\frac{1}{6}$ (B) $\frac{4}{3}$

   (C) $\frac{3}{2}$ (D) $\frac{5}{3}$

5. The area of the region $\{(x, y) : xy \le 8, 1 \le y \le x^2\}$ is **[JEE (Adv)-2019 (Paper-1)]**

   (A) $8 \log_e 2 - \frac{7}{3}$

   (B) $16 \log_e 2 - 6$

   (C) $8 \log_e 2 - \frac{14}{3}$

   (D) $16 \log_e 2 - \frac{14}{3}$

6. Let the functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = e^{x-1} - e^{-|x-1|} \text{ and } g(x) = \frac{1}{2}(e^{x-1} + e^{1-x}).$$

Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is

[JEE (Adv)-2020 (Paper-1)]

(A) $\left(2-\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

(B) $\left(2+\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

(C) $\left(2-\sqrt{3}\right)+\frac{1}{2}\left(e+e^{-1}\right)$

(D) $\left(2+\sqrt{3}\right)+\frac{1}{2}\left(e+e^{-1}\right)$

7. The area of the region

$\left\{(x, y) : 0 \le x \le \frac{9}{4}, \quad 0 \le y \le 1, \quad x \ge 3y, \quad x + y \ge 2\right\}$ is **[JEE (Adv)-2021 (Paper-1)]**

(A) $\frac{11}{32}$

(B) $\frac{35}{96}$

(C) $\frac{37}{96}$

(D) $\frac{13}{32}$

One or More Option(s) Correct Type Questions

8. Let S be the area of the region enclosed by $y = e^{-x^2}$, $y = 0$, $x = 0$, and $x = 1$. Then

[IIT-JEE-2012 (Paper-1)]

(A) $S \ge \frac{1}{e}$

(B) $S \ge 1 - \frac{1}{e}$

(C) $S \le \frac{1}{4}\left(1+\frac{1}{\sqrt{e}}\right)$

(D) $S \le \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}}\left(1-\frac{1}{\sqrt{2}}\right)$

9. If the line $x = \alpha$ divides the area of region $R = \left\{(x,y) \in \mathbb{R}^2 : x^3 \le y \le x, 0 \le x \le 1\right\}$ into two equal parts, then

[JEE (Adv)-2017 (Paper-2)]

(A) $0 < \alpha \le \frac{1}{2}$

(B) $2\alpha^4 - 4\alpha^2 + 1 = 0$

(C) $\alpha^4 + 4\alpha^2 - 1 = 0$

(D) $\frac{1}{2} < \alpha < 1$

Integer / Numerical Value Type Questions

10. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \le d_1(P) + d_2(P) \le 4$, is **[JEE (Adv)-2014 (Paper-1)]**

11. Let $F(x) = \int_x^{x^2+\frac{\pi}{6}} 2\cos^2 t(dt)$ for all $x \in \mathbb{R}$ and $f : \left[0, \frac{1}{2}\right] \to [0, \infty)$ be a continuous function. For $a \in \left[0, \frac{1}{2}\right]$, if $F'(a) + 2$ is the area of the region bounded by $x = 0$, $y = 0$, $y = f(x)$ and $x = a$, then $f(0)$ is

[JEE (Adv)-2015 (Paper-1)]

12. A farmer F_1 has a land in the shape of a triangle with vertices at $P(0, 0)$, $Q(1, 1)$ and $R(2, 0)$. From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n$ $(n > 1)$. If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR, then the value of n is _____. **[JEE (Adv)-2018 (Paper-1)]**

13. Consider the functions f, $g : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2 + \frac{5}{12}$ and $g(x) = \begin{cases} 2\left(1 - \frac{4|x|}{3}\right) &, \ |x| \le \frac{3}{4}, \\ 0 &, \ |x| > \frac{3}{4}. \end{cases}$

If a is the area of the region $\left\{(x,y) \in \mathbb{R} \times \mathbb{R} : |x| \le \frac{3}{4},\ 0 \le y \le \min\{f(x), g(x)\}\right\}$, then the value of $9a$ is ______.

[JEE (Adv)-2022 (Paper-2)]

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